

Cost primal (LP-c)

$$\begin{aligned} \min_p \quad & \sum_g c_g p_g \\ \text{s.t.} \quad & \sum_g p_g = D_t \quad [\lambda_t] \\ & 0 \leq p_g \leq \bar{p}_g \quad [\mu_g] \end{aligned}$$

strong duality

Cost dual (LP-c*)

$$\begin{aligned} \max_{\lambda, \mu} \quad & D_t \lambda - \sum_g \bar{p}_g \mu_g \\ \text{s.t.} \quad & \lambda - \mu_g \leq c_g \quad \forall g \\ \text{KKT:} \quad & \lambda_t = c_m(t) \end{aligned}$$

analogue (swap $c_g \rightarrow e_g$)**Emission primal (LP-e)**

$$\begin{aligned} \min_p \quad & \sum_g e_g p_g \\ \text{s.t.} \quad & \sum_g p_g = D_t \quad [\varepsilon_t] \\ & 0 \leq p_g \leq \bar{p}_g \quad [\nu_g] \end{aligned}$$

strong duality

Emission dual (LP-e*)

$$\begin{aligned} \max_{\varepsilon, \nu} \quad & D_t \varepsilon - \sum_g \bar{p}_g \nu_g \\ \text{s.t.} \quad & \varepsilon - \nu_g \leq e_g \quad \forall g \\ \text{KKT:} \quad & \varepsilon_t = e_m'(t) \end{aligned}$$

add p_{CO_2} adder**Joint primal (LP-j, $p_{\text{CO}_2} > 0$)**

$$\begin{aligned} \min_p \quad & \sum_g (c_g + p_{\text{CO}_2} e_g) p_g \\ \text{s.t.} \quad & \sum_g p_g = D_t \quad [\tilde{\lambda}_t] \end{aligned}$$

strong duality

Joint dual (LP-j*)

$$\begin{aligned} \text{KKT:} \quad & \tilde{\lambda}_t = c_m''(t) + p_{\text{CO}_2} \cdot e_m''(t) \\ \Rightarrow \quad & m''(t) \text{ shifts as } p_{\text{CO}_2} \uparrow \end{aligned}$$